

CODE

```
!pip install transformers accelerate pillow gradio

from transformers import Blip2Processor, Blip2ForConditionalGeneration
from PIL import Image
import torch
import gradio as gr

# Load the model
processor =
Blip2Processor.from_pretrained("Salesforce/blip2-flan-t5-xl")
model = Blip2ForConditionalGeneration.from_pretrained(
    "Salesforce/blip2-flan-t5-xl",
    torch_dtype=torch.float16,
    device_map="auto"
)

# Function to generate description
def describe(image):
    inputs = processor(image, return_tensors="pt").to("cuda")
    output = model.generate(**inputs)
    caption = processor.decode(output[0], skip_special_tokens=True)
    return caption

# Gradio UI
interface = gr.Interface(
    fn=describe,
    inputs=gr.Image(type="pil"),
    outputs="text",
    title="Image Description using VLM (BLIP-2)"
)

interface.launch()
```